

LANGLEY AFB, VA, USA

WMO: 745980

Lat: 37.083N

Lon: 76.360W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13702

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	23.8	0.9	5.8	26.2	5.9	7.4	29.2	26.5	41.9	24.3	44.0	8.8	330	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	92.5	76.3	90.4	75.8	88.3	75.0	79.5	87.0	78.4	85.6	77.4	84.4	10.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	142.9	82.2	76.6	138.8	81.7	75.3	132.7	80.8	42.9	86.8	41.7	85.9	40.7	84.1	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.0	20.4	18.4	DB	13.7	96.4	5.4	2.6	9.8	98.3	6.6	99.8	3.6	101.3	-0.4	103.3
			WB	11.9	81.6	5.0	1.7	8.3	82.8	5.3	83.8	2.5	84.8	-1.2	86.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.1	39.6	42.9	49.0	59.0	67.3	75.4	79.2	77.6	72.5	61.9	51.3	44.6	(6)
	DBStd	15.82	10.46	9.28	9.38	8.72	7.46	5.82	4.57	4.37	5.55	7.74	8.29	9.39	(7)
	HDD50	1050	353	237	137	15	0	0	0	0	0	5	80	223	(8)
	HDD65	3415	787	620	501	225	67	4	0	0	8	153	416	634	(9)
	CDD50	4745	32	39	106	285	535	762	904	856	675	375	120	56	(10)
	CDD65	1632	1	1	5	44	137	316	439	391	233	58	5	2	(11)
	CDH74	13784	1	7	44	351	993	2772	4368	3451	1465	308	18	6	(12)
	CDH80	4559	0	0	3	76	281	988	1687	1118	349	56	1	0	(13)
(14) Wind	WSAvg	9.2	9.7	10.0	10.8	10.6	9.5	8.7	8.3	7.8	8.6	8.4	8.6	8.8	(14)
(15) Precipitation	PrecAvg	46.9	3.7	3.6	4.2	2.9	4.0	4.2	4.9	4.8	4.7	3.1	3.1	3.5	(15)
	PrecMax	68.0	8.7	5.9	10.9	8.4	8.7	10.5	10.6	9.8	15.5	12.0	6.7	7.6	(16)
	PrecMin	28.3	0.7	0.2	1.1	0.5	0.4	0.6	1.2	0.6	0.6	0.4	0.3	0.4	(17)
	PrecStd	10.0	1.5	1.4	2.0	1.7	2.0	2.5	2.4	2.5	3.5	2.2	1.8	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.6	74.3	79.2	86.3	90.2	94.9	96.5	93.4	90.4	85.1	76.6	73.9	(19)
		MCWB	63.2	63.3	62.9	68.3	73.2	75.6	76.5	77.1	74.4	71.7	66.0	64.6	(20)
	2%	DB	65.9	68.4	73.8	81.9	86.3	91.0	93.0	90.9	87.2	81.3	72.1	68.4	(21)
		MCWB	59.5	58.6	60.6	65.7	71.4	74.7	77.2	76.4	74.4	70.0	64.7	62.9	(22)
	5%	DB	61.4	63.7	69.7	78.5	83.2	88.4	90.7	88.5	84.0	77.4	68.1	64.1	(23)
		MCWB	56.6	56.9	58.1	64.5	70.6	74.6	76.5	75.6	73.3	69.0	61.8	60.3	(24)
	10%	DB	56.4	58.8	64.5	73.4	79.8	85.9	88.4	86.1	81.3	73.8	64.2	60.2	(25)
		MCWB	51.5	51.7	57.2	62.9	68.7	73.4	75.9	75.2	72.4	66.9	59.1	56.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.8	65.7	66.5	72.0	76.4	79.1	81.3	81.2	78.8	76.2	69.9	68.3	(27)
		MCDB	68.8	70.6	72.4	79.6	84.6	87.8	89.6	88.7	83.3	80.3	73.0	70.8	(28)
	2%	WB	61.9	62.5	64.0	69.1	74.1	77.8	79.8	79.4	77.5	73.6	66.7	64.3	(29)
		MCDB	65.5	66.6	69.6	77.2	82.1	86.3	88.1	86.2	82.4	78.0	69.9	67.1	(30)
	5%	WB	57.5	58.0	61.7	67.0	72.6	76.6	78.5	78.2	76.2	71.3	63.6	60.9	(31)
		MCDB	60.7	62.6	66.9	74.5	79.7	84.3	86.4	84.8	81.0	75.4	66.0	63.7	(32)
	10%	WB	52.1	52.7	58.0	64.7	70.9	75.4	77.5	77.1	74.8	68.6	60.4	56.5	(33)
		MCDB	56.1	57.8	64.3	72.0	77.2	82.3	85.0	83.3	79.3	72.8	63.9	59.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	17.0	17.0	18.2	16.4	15.5	15.2	14.4	14.2	16.4	17.0	16.0	(35)
		MCDBR	22.6	24.6	24.0	22.8	20.2	18.8	17.8	16.9	16.8	18.7	20.3	20.0	(36)
	5% WB	MCWBR	17.4	17.3	14.1	10.7	9.2	7.4	6.7	6.7	7.2	9.7	13.3	15.8	(37)
		MCWBR	21.4	22.9	20.5	20.3	17.2	15.9	15.6	14.7	12.9	14.6	17.0	19.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.307	0.319	0.346	0.392	0.435	0.485	0.514	0.508	0.409	0.355	0.323	0.311	(40)
		taud	2.542	2.504	2.460	2.341	2.253	2.165	2.098	2.100	2.366	2.516	2.553	2.563	(41)
	Ebn at Noon		282	292	292	282	270	256	247	246	267	275	273	273	(42)
		Edh at Noon	24	29	33	39	43	48	50	49	36	28	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		740	981	1309	1664	1877	2020	1925	1736	1411	1118	837	658	(44)
	RadStd		60	106	104	121	166	133	94	95	118	128	69	52	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.70	N/A	-3.85	N/A	N/A	N/A	N/A	N/A	+277	+193
Regional (61 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+295	+189

Nomenclature: See separate page